Lecture 03.11: Vowel Count (String Traversal & Filtering)

📌 Executive Summary

String character stream filtering, vowel set membership testing in O(1) time, and case normalization.

🔬 Theoretical Foundations & Principles

1. Character Stream Membership Problem

Given a string $S = (c_1, c_2, \ldots, c_N)$ and vowel set $\mathcal{V} = \{'a', 'e', 'i', 'o', 'u'\}$, compute:

$$V(S) = \sum_{i=1}^{N} \mathbb{I}_{\mathcal{V}}(\text{lower}(c_i))$$

2. Data Structure Optimization for Membership

Checking `char in "aeiouAEIOU"` scans a 10-character string linearly ($mathcalO(|$ $mathcalV|)$). Using a `frozenset` performs hash lookups in $mathcalO(1)$ time.

💻 Production Implementation & Code

```
def count_vowels(text: str) -> dict[str, int]:
    vowels = {'a', 'e', 'i', 'o', 'u'}
    counts = {v: 0 for v in vowels}
    for char in text.lower():
        if char in vowels:
            counts[char] += 1
    return counts
```

```
sample = "Artificial Intelligence & Deep Learning 2026"
print("Vowel Frequencies:", count_vowels(sample))
```

⚙️ Algorithmic Complexity & Systems Analysis

- **Time Complexity:** Detailed in respective algorithmic operations.
- **Auxiliary Space:** Memory bounds and CPython frame management.